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An electrostatic master–slave mechanism with force enhancement by external inductors for cancellation of self- and parasitic capacitances

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ABSTRACT

This paper reports a development of an electrostatic master–slave mechanism: a passively synchronized pair of electrostatic motors. The synchronization of the positions of the two motors and the transmission of forces from one motor to the other, are realized by the position dependence of the induced voltages on the electrodes of two sliders. The forces of the mechanism are enhanced by three external inductors. The inductors cancel self- and parasitic capacitances of the slider electrodes, and the cancellation increases the voltages induced on the electrodes. A simple model is provided to explain the principles of the mechanism and the theory of the force enhancement. A prototype using a pair of three-layered electrostatic film motors was fabricated. The synchronization of the two motors with a velocity of 20 mm/s and an acceleration of 0.16 m/s^2 was demonstrated under the input voltage of 660 V/5.95 kHz, and the maximum force that could be transmitted was 0.6 N.

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1. Introduction

Electrostatic motors had long been assumed less powerful compared to electromagnetic motors. This assumption was proved to be wrong in the microscopic region by the revolutionary advance in the field of MEMS (microelectromechanical systems) [1,2]. Since then, various types of electrostatic motors were developed and studied, including variable capacitance motors [3–6], induction motors [7], electret motors [8] and corona motors [9]. These electrostatic motors provide means of actuation for small-sized integrated systems such as portable electronic devices.

On the other hand, there is a different approach in utilizing the scaling law of the electrostatic forces. The approach is to integrate microscopic motor elements in two- and three-dimensional manner, to obtain larger forces that can even be utilized in macro-scale applications of centimeter or larger scale. Such an idea was demonstrated by Niino et al. in the surface drive type electrostatic film motor named DEMED (dual excitation multi-phase electrostatic drive) [10]. A maximum power-to-weight ratio of 230 W/kg [11] and a thrust force of 310 N [12] was demonstrated with this motor, which shows enough performance to replace electromagnetic motors in macro-scale applications.

DEMED is a promising motor for applications in special environments where non-magnetic characteristics are required. One such example is inside MRI (Magnetic Resonance Imaging) devices. MRI devices require the equipments that are used inside them to be constructed with non-magnetic materials, and also require them to operate without producing magnetic fields. DEMED satisfies both requirements, and its MR compatibility was proved in [13]. Another example of such an environment is inside the vacuum chambers of the equipments whose process involves electron beams. These equipments include electron beam lithography systems and scanning electron microscopes. For these applications, an electrostatic levitation motor based on the DEMED technology was proposed [14].

To provide a simple man–machine interface for the above mentioned environments, we have previously reported an electrostatic master–slave mechanism using a pair of DEMED type electrostatic motors [15]. The mechanism enables passive synchronization of the two motors by electrostatic inductions: the positions of the two motors are synchronized and the forces of the motors are transmitted from one to the other. Analogous systems using the electromagnetic motors are called synchros (also known as selsyn machines) [16]. Synchros are well-established and have long been utilized as sensors or power transmitters. The electrostatic master–slave mechanism can be used for the same purposes, with advantages of electrostatic motors: light weight, thin structure, non-magnetic and low heat dissipation. The mechanism opens a way to applications such as remote manipulations of objects from outside the vacuum chamber and a communication between two people in and out of the MRI. These concepts are shown in Fig. 1.

However, the previous prototype of the electrostatic master–slave mechanism was found to be insufficient for practical

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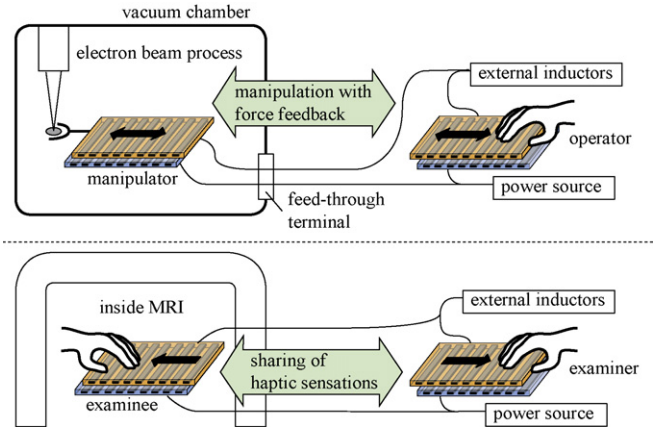


Fig. 1. Expected applications of the electrostatic master–slave mechanism.

use. The forces it could generate were small: they were barely enough to overcome the friction and to perform quasi-static synchronization with a maximum velocity of 0.1 mm/s. In this paper, we introduce a method to enhance the forces. It is found that the forces are lowered by self- and parasitic capacitances of the slider electrodes, which reduce the voltages induced on the electrodes. In the force enhancement method, the self- and the parasitic capacitances are canceled with externally added inductors. Section 2 focuses on the modeling of the mechanism, and simple equations to explain the principles are provided. Section 3 introduces the concept of the force enhancement method, based on the analysis of the derived model. Finally, the results of the experiments are shown in Section 4, which verify the theory and show the improved performance of the mechanism, followed by the discussions in Section 5.

2. Principle of the electrostatic master–slave mechanism

This section focuses on a modeling of the electrostatic master–slave mechanism. The model is a pair of six-terminal capacitance networks connected in parallel, which are idealized under some assumptions. The idealization provides simple equations that express the two important characteristics of the electrostatic master–slave mechanism: the synchronization of positions and the transmission of forces.

A schematic of the electrostatic master–slave mechanism is shown in Fig. 2. The mechanism consists of two DEMED type electrostatic motors connected in parallel, and two voltage sources.

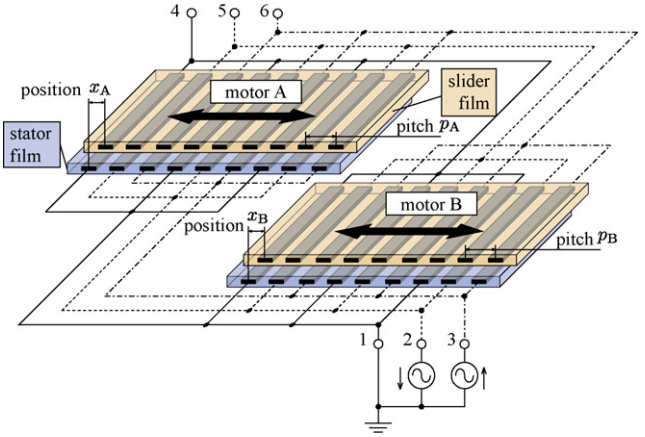


Fig. 2. Schematic of the electrostatic master–slave mechanism.

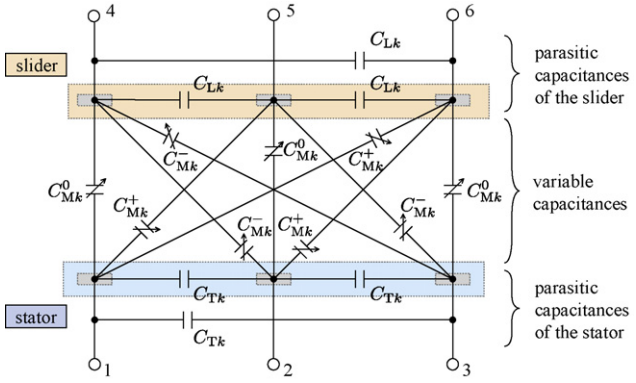


Fig. 3. The six-terminal capacitance network model of the electrostatic motor *k*.

The voltage sources have same amplitudes but inverted phases. In previous studies, the DEMED type motors were operated by three-phase ac voltages, applied to both the slider and the stator. For the electrostatic master–slave mechanism, required number of voltage sources is two, and the voltages are applied only to the stator side. In principle, the voltage sources can be dc voltages with the same magnitude but different polarities. However, as pointed out in [17], the thrust force of DEMED type motors are lowered when they are operated under dc voltages. Therefore, ac voltage sources are used instead of dc voltages.

The two motors are distinguished by the names A and B. The electrodes of each motor are connected to their counterparts. The connected sets of electrodes are referred to by the phase numbers 1 through 6, as shown in Fig. 2. The phase 1 in the stator side is connected to the ground, whereas the phase 2 and phase 3 are connected to the two ac voltage sources.

2.1. Idealized model of a single motor

The two motors A and B are of the same type, therefore the same explanation applies. The following explanations are generalized on the motor *k*. The symbol *k* is to be replaced by A or B for the corresponding motor.

The motor *k* is made up of two films: a stator film and a slider film. In each film, three-phase parallel electrodes are embedded with a constant pitch p_k. The actuated degree-of-freedom is the slider position x_k, which is in the direction along the electrode surface, as shown in Fig. 2. For the simplicity of expressions, the positions are also represented by angles $\theta_k = (2\pi/3p_k)x_k$. This representation is useful because the geometry of the motor is periodic with a period of three electrode pitches.

A six-terminal capacitance network model, which was proposed in [18], is defined for the motor *k* as shown in Fig. 3. In the model, the distributed capacitances of the electrodes are modeled as lumped capacitance elements. To distinguish them clearly, we define the following names for the capacitances. The capacitances connecting two phases from the same film are defined as the *parasitic capacitances*. The rest of the capacitances that connect two phases from the different films, one from the stator and the other from the slider, are defined as the *variable capacitances*. The parasitic capacitances are represented by the symbol C_{Tk} and C_{Lk}, respectively, for the stator and the slider. Capacitances with the subscript M represent the variable capacitances.

For the simplicity of analysis, we make the following assumptions to idealize the model.

1. The three phases within the same film are geometrically symmetric. Therefore the capacitances of the three phases are also symmetric.

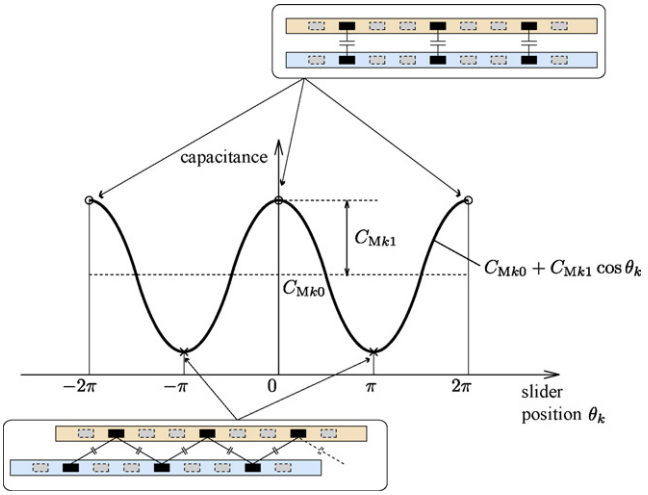


Fig. 4. Schematic of an idealized variable capacitance. The capacitance is given by the cosine of the slider position represented in an angle θ_k, with an amplitude of C_{Mk1} and an offset of C_{Mk0}. Note that $C_{Mk1} \leq C_{Mk0}$, because negative capacitance does not exist.

2. The variable capacitances are given as biased cosine functions of the slider position, as shown in Fig. 4.
3. The parasitic capacitances are independent of the slider positions.
4. The capacitances formed between each phase and the ground are small enough to be neglected.

Under these assumptions, the parasitic capacitances C_{Tk} and C_{Lk} are constant, and the variable capacitances are defined as

$$\begin{cases} C^0_{Mk} = C_{Mk0} + C_{Mk1}\cos\theta_k \\ C^+_{Mk} = C_{Mk0} + C_{Mk1}\cos\left(\theta_k + \frac{2}{3}\pi\right) \\ C^-_{Mk} = C_{Mk0} + C_{Mk1}\cos\left(\theta_k - \frac{2}{3}\pi\right) \end{cases}. \tag{1}$$

A six-by-six capacitance coefficient matrix $\mathbf{C}_k$ is defined for this model. The matrix is a set of coefficients representing the voltage-to-charge relationships within the six phases. The relationships are written as

$$\begin{bmatrix} \mathbf{q}_{STk} \\ \mathbf{q}_{SLk} \end{bmatrix} = \mathbf{C}_k \begin{bmatrix} \mathbf{v}_{STk} \\ \mathbf{v}_{SLk} \end{bmatrix}, \tag{2}$$

where $\mathbf{q}_{STk}$ and $\mathbf{q}_{SLk}$ are the charge vectors, and $\mathbf{v}_{STk}$ and $\mathbf{v}_{SLk}$ are the voltage vectors, respectively, defined as

$$\begin{aligned} \mathbf{q}_{STk} &= {}^{t}\begin{bmatrix} q_{1k} & q_{2k} & q_{3k} \end{bmatrix}, \\ \mathbf{q}_{SLk} &= {}^{t}\begin{bmatrix} q_{4k} & q_{5k} & q_{6k}, \end{bmatrix}, \\ \mathbf{v}_{STk} &= {}^{t}\begin{bmatrix} v_{1k} & v_{2k} & v_{3k} \end{bmatrix} \text{ and} \\ \mathbf{v}_{SLk} &= {}^{t}\begin{bmatrix} v_{4k} & v_{5k} & v_{6k} \end{bmatrix}. \end{aligned} \tag{3}$$

The numbers 1 through 6 in the subscripts of the elements indicate that the voltage (or the charge) is concerning the corresponding phase. The vectors are expressed in three-dimensional column vectors instead of six, because we want to treat the stator phases and the slider phases separately. For the same reason, we define the matrix $\mathbf{C}_k$ in the two-by-two block matrix form:

$$\mathbf{C}_k = \begin{bmatrix} \mathbf{C}_{STk} & \mathbf{C}_{Mk} \\ {}^{t}\mathbf{C}_{Mk} & \mathbf{C}_{SLk} \end{bmatrix}, \tag{4}$$

where each block is defined as

$$\mathbf{C}_{STk} = \begin{bmatrix} C_{STk} & -C_{Tk} & -C_{Tk} \\ -C_{Tk} & C_{STk} & -C_{Tk} \\ -C_{Tk} & -C_{Tk} & C_{STk} \end{bmatrix}, \tag{5}$$

$$\mathbf{C}_{SLk} = \begin{bmatrix} C_{SLk} & -C_{Lk} & -C_{Lk} \\ -C_{Lk} & C_{SLk} & -C_{Lk} \\ -C_{Lk} & -C_{Lk} & C_{SLk} \end{bmatrix}, \tag{6}$$

and

$$\mathbf{C}_{Mk} = \begin{bmatrix} -C^0_{Mk} & -C^+_{Mk} & -C^-_{Mk} \\ -C^-_{Mk} & -C^0_{Mk} & -C^+_{Mk} \\ -C^+_{Mk} & -C^-_{Mk} & -C^0_{Mk} \end{bmatrix}. \tag{7}$$

The capacitance coefficient C_{STk} is the *s* elf-capacitance of a phase in the stator, and C_{SLk} is the same for the slider. The self-capacitance of a phase is defined as the sum of all the capacitances connected to the phase. Therefore, they are defined as:

$$\begin{cases} C_{STk} = 3C_{Mk0} + 2C_{Tk} \\ C_{SLk} = 3C_{Mk0} + 2C_{Lk} \end{cases}. \tag{8}$$

2.2. Position dependency of the slider voltages

To obtain the forces of the two motors, we must first know the voltages. The stator voltages are given because the stator phases are connected to either the ground or a voltage source, whereas the slider voltages are unknown. To solve for the slider voltages, we derive differential equations by the following procedure.

The voltage vectors of the two motors are equal, because they are connected in parallel. Here we define the common voltage vector of the stator side $\mathbf{v}_{ST}$ as

$$\mathbf{v}_{ST} = {}^{t}[0 \quad -v_0\cos\omega t \quad v_0\cos\omega t], \tag{9}$$

where v_0 is the voltage amplitude and ω is the frequency of the two voltage sources. The voltage of the phase 3, $v_0\cos\omega t$, will be referred to as the *input voltage*. The common voltage vector of the slider side $\mathbf{v}_{SL}$ is defined as

$$\mathbf{v}_{SL} = {}^{t}[v_4 \quad v_5 \quad v_6]. \tag{10}$$

where v_4, v_5 and v_6 are the slider voltages of the phase specified by their subscripts.

The charge vectors of the slider side of each motor are then expressed as

$$\mathbf{q}_{SLk} = {}^{t}\mathbf{C}_{Mk}\mathbf{v}_{ST} + \mathbf{C}_{SLk}\mathbf{v}_{SL}, \quad (k = \text{A,B}). \tag{11}$$

To derive an equilibrium equation, the charges are converted into current representations. Using the dot notation of the first time derivatives, the currents are written as

$$\dot{\mathbf{q}}_{SLk} = \dot{\theta}_k \frac{\partial {}^{t}\mathbf{C}_{Mk}}{\partial \theta_k}\mathbf{v}_{ST} + {}^{t}\mathbf{C}_{Mk}\dot{\mathbf{v}}_{ST} + \mathbf{C}_{SLk}\dot{\mathbf{v}}_{SL}, \quad (k = \text{A,B}). \tag{12}$$

The currents are positive when they flow into the capacitances. Note that the first time derivative of the slider positions (expressed in an angular form) appear in the coefficient of the first term, because the variable capacitances are implicit functions of time when the motors are moving.

From Kirchhoff's current law, the sum of the two current vectors $\dot{\mathbf{q}}_{SLA}$ and $\dot{\mathbf{q}}_{SLB}$ equals zero. Thus, the equilibrium equation is written as

$$\mathbf{0} = \sum_{k=\text{A,B}} \left(\dot{\theta}_k \frac{\partial {}^{t}\mathbf{C}_{Mk}}{\partial \theta_k}\mathbf{v}_{ST} + {}^{t}\mathbf{C}_{Mk}\dot{\mathbf{v}}_{ST} + \mathbf{C}_{SLk}\dot{\mathbf{v}}_{SL} \right). \tag{13}$$

The differential equation can be solved for $\mathbf{v}_{SL}$ as

$$\mathbf{v}_{SL} = -\frac{\sqrt{3}v_0}{C_P}\cos\omega t \sum_{k=A,B} \begin{bmatrix} C_{Mk}\sin\theta_k \\ C_{Mk}\sin\left(\theta_k + \frac{2\pi}{3}\right) \\ C_{Mk}\sin\left(\theta_k - \frac{2\pi}{3}\right) \end{bmatrix}, \tag{14}$$

where C_P is the sum of the self- and the parasitic capacitances of the sliders, which is defined as

$$C_P = C_{SLA} + C_{SLB} + C_{LA} + C_{LB}. \tag{15}$$

To simplify the expression, we propose the following representation of the slider voltages:

$$\mathbf{v}_{SL} = v_i \cos(\omega t)\mathbf{r}, \tag{16}$$

where v_i is the maximum amplitude of the slider voltages, and $\mathbf{r}$ is the normalized position dependence vector defined as

$$v_i = \frac{\sqrt{3}(C_{MA1} + C_{MB1})}{C_P} v_0 \tag{17}$$

and

$$\mathbf{r} = -\frac{1}{C_{MA1} + C_{MB1}} \sum_{k=A,B} \begin{bmatrix} C_{Mk}\sin\theta_k \\ C_{Mk}\sin\left(\theta_k + \frac{2\pi}{3}\right) \\ C_{Mk}\sin\left(\theta_k - \frac{2\pi}{3}\right) \end{bmatrix}. \tag{18}$$

The position dependence vector $\mathbf{r}$ is defined so that the values of its elements change from −1 to 1, depending on the two slider positions. Therefore, the slider voltages are represented by the single voltage amplitude v_i.

2.3. Forces generated by the two motors

The electrostatic force generated by each motor can be calculated based on the principle of virtual work. The force of the motor A, f_A is calculated as

$$f_A = \frac{1}{2}\left[{}^t\mathbf{v}_{ST} \quad {}^t\mathbf{v}_{SL}\right]\frac{\partial \mathbf{C}_A}{\partial x_A}\begin{bmatrix}\mathbf{v}_{ST} \\ \mathbf{v}_{SL}\end{bmatrix} = \frac{\kappa v_0 v_i}{p_A}\sin\left(\theta_A - \theta_B\right)(1 + \cos 2\omega t), \tag{19}$$

where κ is a constant defined as

$$\kappa = \frac{\sqrt{3}\pi}{2}\frac{C_{MA1}C_{MB1}}{C_{MA1} + C_{MB1}}. \tag{20}$$

The expression (19) indicates that the force oscillates with a frequency of 2ω. However, the oscillation is damped and can be neglected when the frequency is set higher than the mechanical response of the system: in the order of kHz or higher [17]. In such case, only the offset of the force is effective. Therefore, the effective force of the motor A, $\bar{f}_A$ is obtained as

$$\bar{f}_A = -\frac{\kappa v_0 v_i}{p_A}\sin\left(\theta_A - \theta_B \mp \pi\right). \tag{21}$$

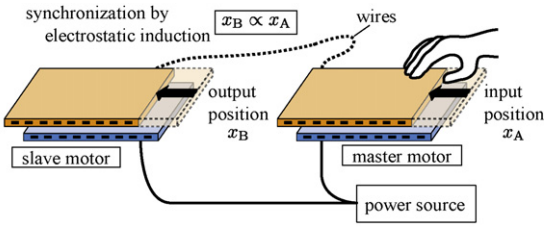
Note that the phase offset of $\mp\pi$ is intentionally added so that the minus sign appears in the coefficient, to indicate that the force is a restoring force about $\theta_A = \theta_B \pm \pi$.

The effective force of motor B, $\bar{f}_B$ can be calculated by the same procedure as

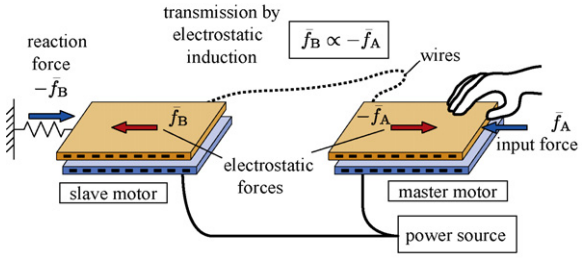
$$\bar{f}_B = -\frac{\kappa v_0 v_i}{p_B}\sin\left(\theta_B - \theta_A \pm \pi\right), \tag{22}$$

which is also a restoring force, about $\theta_B = \theta_A \mp \pi$.

The two expressions of forces (21) and (22) lead us to two important characteristics of the electrostatic master–slave mechanism.



(a) Synchronization of two motors under no load.



(b) Transmission of forces from one motor to the other.

Fig. 5. Illustration of the two characteristics of the electrostatic master–slave mechanism: (a) synchronization of two motors under no load and (b) transmission of forces from one motor to the other.

1. The two slider positions are synchronized when no external forces are applied. The force $\bar{f}_A$ is a restoring force about $\theta_A = \theta_B \pm \pi$, and $\bar{f}_B$ is a restoring force about $\theta_B = \theta_A \mp \pi$. Therefore, under no load, the two sliders will rest in their positions with position difference of π (in angular representation). The equilibrium condition can also be written as $x_A = (p_A/p_B)x_B \pm (3/2)p_A$, in terms of two slider positions.
2. The magnitudes of the forces generated by the two motors are proportional to each other with a ratio determined by their electrode pitch. Regardless of the position difference, the relationship between the two forces is given as $\bar{f}_A = (-p_B/p_A)\bar{f}_B$. This means that when an external force is applied to one of the two sliders, the force is transmitted to the other slider.

The two characteristics are illustrated in Fig. 5.

These ideas can also be explained by a change of potential distributions of the two sliders. The idealized force characteristics of the two motors are shown in Fig. 6. Standing potential waves are formed on the surface of the two stators by the voltages applied from the voltage sources. On the other hand, the waves formed by the slider voltages are dependent on the positions of the two sliders. At equilibrium points where the position difference is equal to $\pm\pi$ in angular representation, the potential waves are minimized and no forces are generated. For example, when the two motors are identical, all three voltages on the sliders become zero. As one slider is displaced from the equilibrium position, the slider waves increase, and the restoring forces are generated. The equilibrium positions of the two motors are mutually dependent, and thus the synchronization of the two motors is made possible.

3. Force enhancement by cancellation of self- and parasitic capacitances

In this section, a method to enhance the forces of the electrostatic master–slave mechanism is introduced. It is found that the forces are greatly reduced by self- and parasitic capacitances of the slider electrodes. The forces can be enhanced by canceling these capacitances with three external inductors.

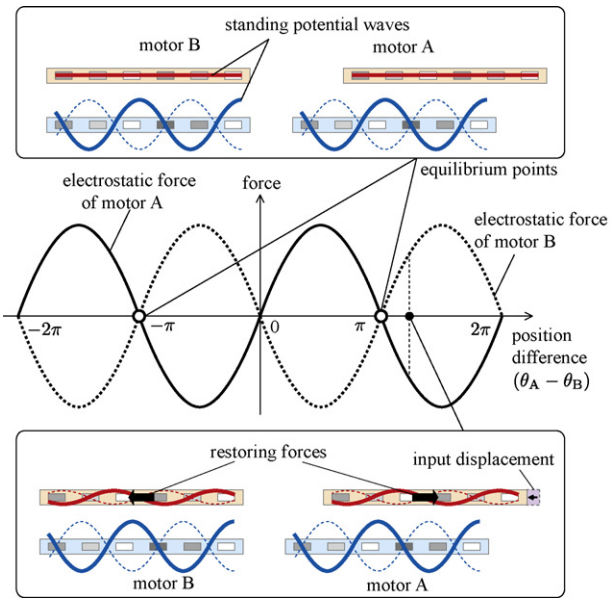


Fig. 6. Illustration of the idealized force characteristics of the electrostatic master–slave mechanism.

3.1. Concept of the self- and the parasitic capacitance cancellation

The forces generated by the master–slave mechanism is limited by the slider voltages. The equations (21) and (22) indicate that the forces are proportional to the two voltage amplitudes: the amplitude of the input voltage v_0 and the maximum amplitude of the slider voltages v_i. The input voltage is set as high as possible under the breakdown strength of the electrodes. On the other hand, the slider voltages are limited to be lower than the input voltage. To obtain larger forces, the slider voltages must be increased up to the same level as the input voltages.

The limitation of the slider voltages is due to the self- and parasitic capacitances of the slider electrodes. From the definition of the self-capacitances in (8), v_i is rewritten as

$$v_i = \frac{C_{MA1} + C_{MB1}}{\sqrt{3}\,(C_{MA0} + C_{MB0} + C_{LA} + C_{LB})} v_0 \le \frac{v_0}{\sqrt{3}}. \tag{23}$$

The voltage is maximized in the most ideal case, where the offsets of the variable capacitances are equal to their amplitudes, and the parasitic capacitances are zero. This indicates that the maximum voltage that can be induced on the slider electrodes is about 58% of the input voltage. The voltage is further reduced from this maximum value, as the self- and the parasitic capacitances of the slider electrodes increase. For example, the corresponding capacitance parameters for the previous prototype in [15] were $C_{Mk1} = 81$ pF, $C_{Mk0} = 126$ pF and $C_{Lk} = 443$ pF (k = A, B), and v_i was calculated to be only 8% of the input voltage. Therefore, the self- and the parasitic capacitances must be reduced to increase the slider voltages.

To reduce the parasitic capacitances, there are two possible methods in terms of electrode design, but they are unlikely to provide drastic increase in the slider voltages. The first method is the use of materials with low dielectric constant as an insulator. For example, benzocyclobutene polymer, a material with low dielectric constant of 2.65, was studied by Modafe et al. to reduce the parasitic capacitances of the electrostatic micromotors [4,19–21]. Another method is to optimize the geometry of the electrodes to have less parasitic capacitances. Examples of the parameters to be optimized are the length of the gap, the space between electrodes and the electrode thickness. These modifications will reduce the parasitic capacitances, but drastic reduction is difficult. Furthermore, the already mentioned limit of 58% cannot be exceeded by these methods.

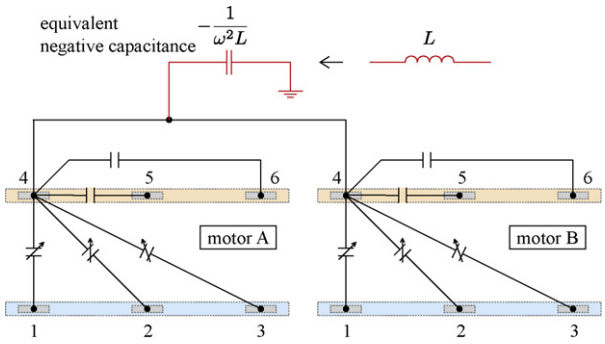


Fig. 7. Illustration of the cancellation of self-capacitance of the phase 4. Only the capacitances involving the self-capacitance of the phase 4 are shown. In the sinusoidal steady-state operation, the self-capacitance is reduced by the equivalent negative capacity $-(1/\omega^2 L)$.

To increase the slider voltages more effectively, we propose the method of canceling the self- and the parasitic capacitances by adding external inductors. The concept of this method is shown in Fig. 7. An inductance L can be modeled as a capacitor with a negative capacitance $-(1/\omega^2 L)$ in sinusoidal steady-state operation at frequency ω. When this negative capacitance element is connected between a slider phase and the ground, the negative capacitance is added to the self-capacitance of the phase. The sum of the self- and the parasitic capacitances C_P is then reduced to $C_P - (1/\omega^2 L)$, and the slider voltages are increased as

$$v_i = \frac{\sqrt{3}\,(C_{MA1} + C_{MB1})}{C_P - (1/\omega^2 L)} v_0. \tag{24}$$

The negative capacitance reduces the self- and the parasitic capacitances of the slider electrodes as a whole, and the 58% limit does not apply. Therefore, this method enables the slider voltages to increase up to the same level as the input voltages.

3.2. Modeling of the electrostatic master–slave mechanism with three inductors

In practice, ideal inductors are not available, and therefore the losses of the inductors must also be considered. The losses damp the voltage amplitudes and cause the phases to shift, which have a great influence on the forces as explained later in this section. Figure 8 shows the schematic of the master–slave mechanism with three additional inductors. Each of the three inductors is modeled as a series of an inductance L and a resistance R_S. If we express the currents that flows into the three inductors by three-dimensional column vector $\dot{\mathbf{q}}_C$, the slider voltages are given as

$$\mathbf{v}_{SL} = L\ddot{\mathbf{q}}_C + R_S\dot{\mathbf{q}}_C. \tag{25}$$

From Kirchhoff's current law, the sum of the three current vectors $\dot{\mathbf{q}}_C$, $\dot{\mathbf{q}}_{SLA}$ and $\dot{\mathbf{q}}_{SLB}$ is equal to zero:

$$\mathbf{0} = \dot{\mathbf{q}}_C + \sum_{k=A,B} \left(\dot{\theta}_k \frac{\partial^t \mathbf{C}_{Mk}}{\partial \theta_k} \mathbf{v}_{ST} + {}^t\mathbf{C}_{Mk}\dot{\mathbf{v}}_{ST} + \mathbf{C}_{SLk}\dot{\mathbf{v}}_{SL} \right). \tag{26}$$

The differential equations are nonlinear and cannot be solved analytically. Here we linearize the equations by assuming mechanically quasi-static conditions, where the slider positions are approximated as constant. This assumption is made because our first goal is to derive simple expressions of the forces.

The linearized differential equations can be solved for the slider voltages $\mathbf{v}_{SL}$ as

$$\mathbf{v}_{SL} = v_i' \cos(\omega t + \phi)\mathbf{r}, \tag{27}$$

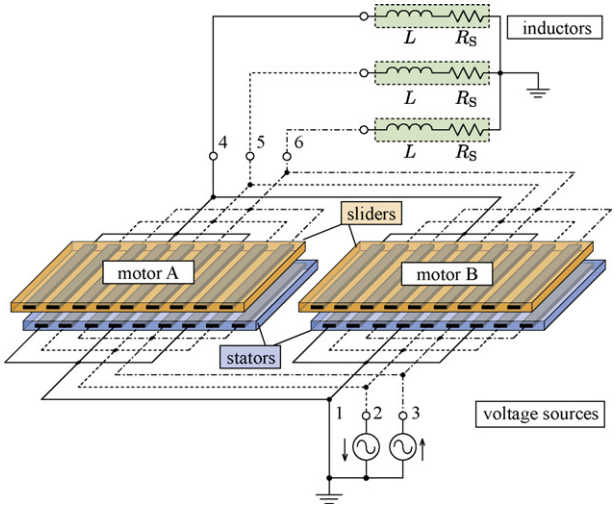


Fig. 8. Schematic of the electrostatic master–slave mechanism with three additional inductors. Each inductor is modeled as a series of an inductance L and a resistance R_S.

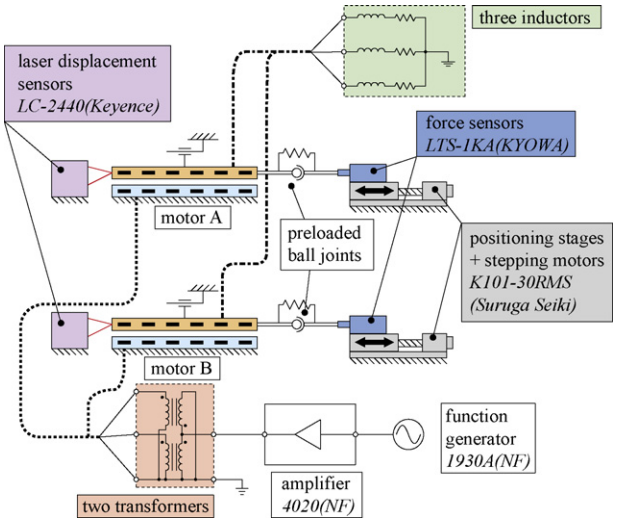


Fig. 9. Schematic diagram of the experimental setup.

where v_i' is the magnitude and ϕ is the argument of the complex function $V_i'(j\omega)$, which is defined as

$$V_i'(j\omega) = \frac{\sqrt{3}\,(C_{MA1} + C_{MB1})}{C_P} v_0 \frac{\omega^2 - 2j\zeta\omega_n\omega}{\omega^2 - 2j\zeta\omega_n\omega - \omega_n^2}. \tag{28}$$

The resonance frequency ω_n and the damping factor ζ are defined as

$$\omega_n = \frac{1}{\sqrt{LC_P}}, \quad \zeta = \frac{R_S}{2}\sqrt{\frac{C_P}{L}}. \tag{29}$$

By the same procedure as (21), the effective forces generated by the two motors are calculated as

$$\bar{f}_A = -\frac{\kappa v_0 v_i'}{p_A}\cos\phi\sin\left(\theta_A - \theta_B \mp \pi\right) \tag{30}$$

and

$$\bar{f}_B = -\frac{\kappa v_0 v_i'}{p_B}\cos\phi\sin\left(\theta_B - \theta_A \pm \pi\right). \tag{31}$$

The forces are proportional to $v_i'\cos\phi$: the real part of the complex function $V_i'(j\omega)$. Therefore, to optimize the forces, the frequency of the input voltage should be decided considering both the amplitude and the phase shift of the slider voltages. When the losses of the inductors are small enough and $\zeta << 1$, the optimum operation frequency is $\omega_{opt} = \omega_n\sqrt{2\zeta + 1}$, and $v_i'\cos\phi$ at this condition is approximated as

$$v_i'\cos\phi = v_i\frac{(2\zeta+1)^2}{4\zeta(\zeta+1)} \simeq \frac{v_i}{4\zeta}, \tag{32}$$

which means that the forces are increased as well by the ratio $1/4\zeta$.

It is also important to note that the equilibrium condition changes depending on the sign of $\cos\phi$. When $\cos\phi > 0$, the equilibrium condition is the same as the case without the inductors, but when $\cos\phi < 0$, the signs of the coefficients in (30) and (31) become positive, and the equilibrium condition becomes $\theta_A = \theta_B$. In other words, stable and unstable zero-crossing points of the forces alter.

4. Experiments

In this section, results of the experiments are described. The first part of the section describes the experimental setup. The second part is on the measurement of the slider voltages, with and without the inductors. The third part describes the measurement of the frequency dependent force characteristics. Finally, the synchronization of the two sliders is demonstrated.

4.1. Experimental setup

Figure 9 shows the schematic diagram of the experimental setup. The setup consists of the two motors, the two voltage sources, the three external inductors and the sensors for force and position measurements.

The configuration of the electrostatic motor unit is shown in Fig. 10. Two units of the same design were used as the motor A and the motor B. The unit consists of three pairs of films stacked in layers, two linear guides, and supporting mechanical parts: a base plate and a slider plate. The reason for using three layers was to have a large force with respect to the measurement range of the force sensors. The dimensions for each pair of films are shown in Fig. 11. Glass beads were scattered in the gap between the films. The function of the glass beads was to maintain a constant gap and also to reduce friction. The gap was filled with dielectric liquid to prevent corona discharges.

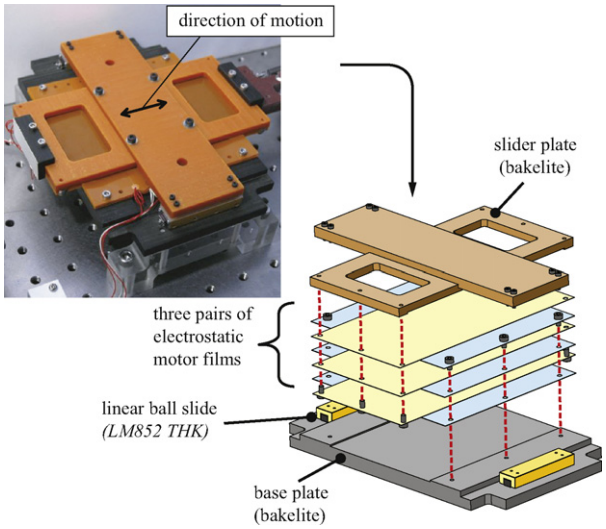


Fig. 10. Configuration of the electrostatic motor unit.

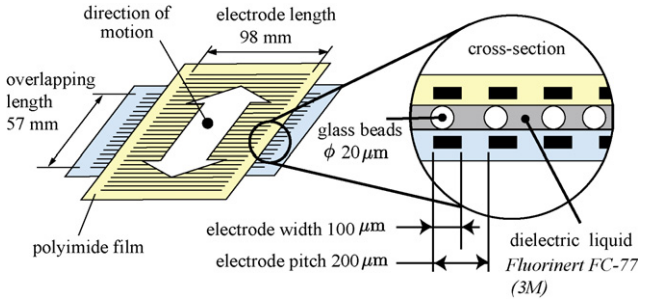


Fig. 11. Dimensions of a single pair of films.

The voltage sources consisted of three parts: a function generator, an amplifier and two transformers. The voltage from the function generator was first amplified 20 times by the amplifier and then amplified 10 times by the two transformers. The two transformers were designed to minimize the reactive current around 5.8 kHz. The ferrite cores used for the transformers were *P* Q5050 (TDK). The number of windings for the primary coil was 22 turns and for the secondary coil was 220 turns. For the details on the design of transformers, see [22]. The connection of the input and the output terminals of the two transformers were determined so that the phases of the two output voltages are inverted.

Three air solenoid coils were used as the external inductors, which were connected to the three phases of the slider side. Specification of the coil is shown in Fig. 12.

For the measurements, laser displacement sensors and force sensors (strain gauges) were installed in the setup. Each of the force sensors was mounted on a linear positioning stage, which was actuated by a stepping motor and a ball screw. The sensing end of the force sensor was connected to the slider of the motor unit, through a preloaded ball joint.

The capacitance parameters of the two motors were identified by the method introduced in [18]. The measured parameters are shown in Fig. 13. The capacitances of the two motors differed, due to the misalignments between the three layers. For both motors, the self-capacitances of the phase 5 was larger than the phases 4 and 6, by about 150 pF. The difference comes from the design of feeding electrodes, as explained in [18]. Capacitors with 307 pF capacitance were added in parallel with the inductors connected to the phases 4 and 6, to compensate for the difference: to have equal resonance frequencies for the three phases.

From the identified capacitance parameters of the motors and the specification of the inductors, resonance frequency was calculated to be $\omega_n = 5.78$ kHz and the damping factor was $\zeta = 0.031$. Therefore, the optimum operation frequency was expected to be $\omega_{opt} = 5.96$ kHz, and the forces are expected to increase by $1/(4\zeta) \approx 8$ times by the added inductors.

4.2. Voltage measurements

First, to show the position dependency of the slider voltages, the voltages were measured with different slider positions, without the external inductors. The slider B was fixed at three positions −150 μm, 50 μm and 250 μm. For each condition, the slider A was moved from −300 μm to 300 μm with an interval of 20 μm. The results are shown in Fig. 14, together with calculated values . Theoretical equilibrium positions of the slider A are also shown in the figure. The slider voltages are minimized at the equilibrium positions as expected. A change in the equilibrium position of the slider A, depending the position of the slider B, is also observed from the result.

Second, the three inductors (and the two capacitors for compensation) were connected to the sliders, and the voltage of the phase 4 was measured under different frequencies. The two sliders were fixed at $x_A = -150$ μm and $x_B = -150$ μm. These positions were chosen because the voltage amplitude and the phase shift measured at these positions correspond to v_i' and ϕ in (27). The results are shown in Fig. 15. The voltages measured without the inductors are also shown in the figure for comparison. The results agree well with the theory.

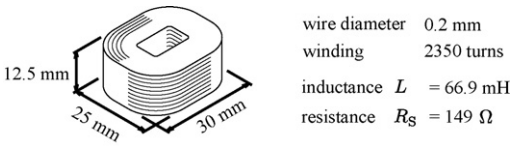


Fig. 12. Specification of the air solenoid coil. The inductance and the resistance was measured using LCZ meter *ZM2372(NF Co.)*, at 1 V_{p-p}/5.8 kHz.

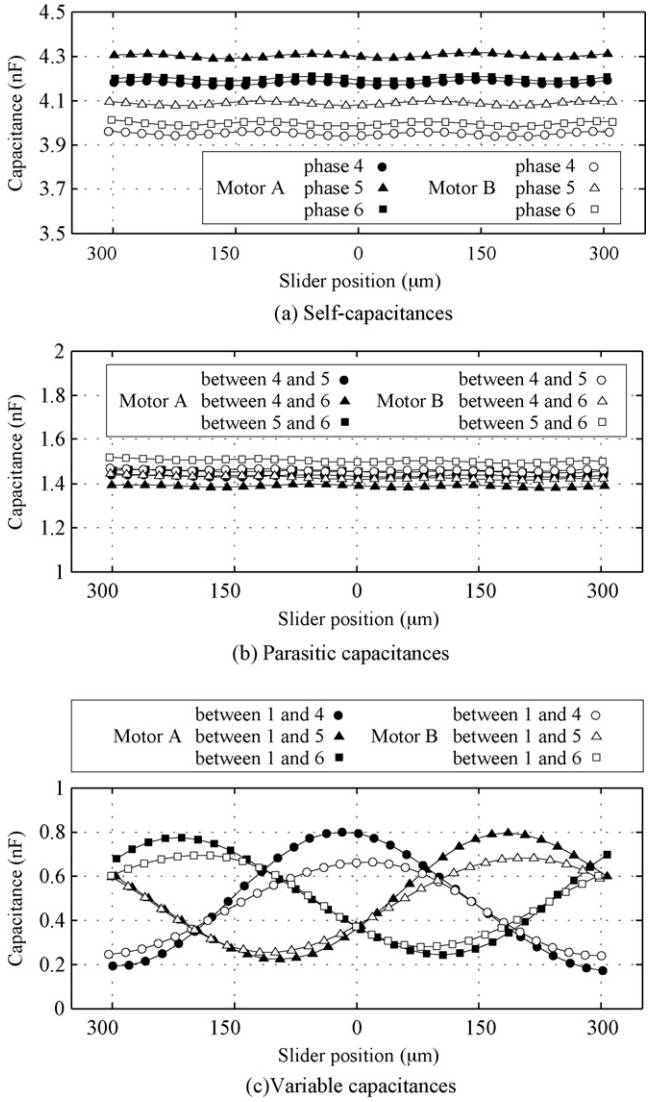


Fig. 13. Measured capacitance parameters of the two motors. Only the parameters involved in the determination of the slider voltages are shown. Variable capacitances involving the phase 2 and the phase 3 are omitted, because they were almost equivalent to that of the phase 1. The measurements were carried out using LCZ meter *ZM2372(NF Co.)*, at 1 V_{p-p}/5.8 kHz: (a) self-capacitances, (b) parasitic capacitances, and (c) variable capacitances.

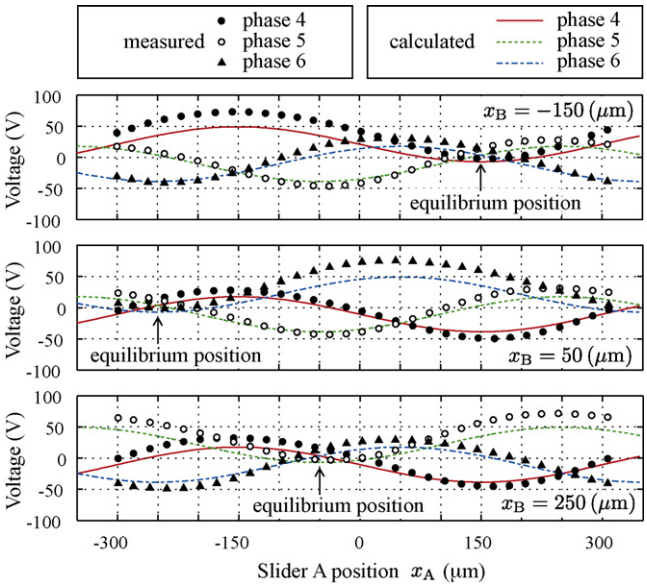


Fig. 14. Position dependent amplitudes of the slider voltages, measured without the external inductors. Negative values indicate that the phase of the voltage is shifted by 180 degrees from the input voltage. The input voltage was 660 V/5.95 kHz. Theoretical equilibrium positions of the slider A are shown by arrows.

4.3. Force measurements

The forces of the two motors were measured by the following procedure. The procedure was necessary to separate the measured forces into two components: the electrostatic thrust force and the friction force.

First, the two sliders were forced to scan the trajectory shown in Fig. 16, by the positioning stages attached to them. During the scanning, the reaction forces from the two sliders were measured by the force sensors mounted on the positioning stages. Second, the force data are filtered digitally with a cut-off frequency of 100 Hz to eliminate the electrical noise. Third, the force data within the range (a) and (b) shown in Fig. 16 were extracted, to eliminate the influence of the acceleration and the deceleration. Finally, the acquired force data are separated into the electrostatic thrust force and the friction

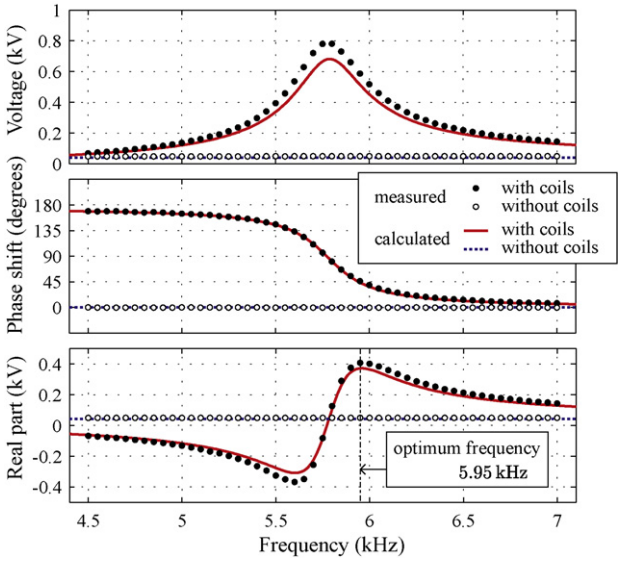


Fig. 15. Frequency dependence of the slider voltage, represented by the voltage of the phase 4 at $(x_A, x_B) = (-150, -150)$ μm. The amplitude of the input voltage was $v_0 = 558$ V.

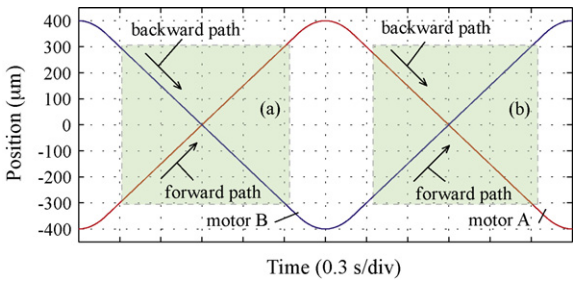


Fig. 16. Scan trajectory of the two motors during the force measurement. The two motors A and B are forced by the positioning stages to follow the trajectory at a constant velocity of 500 μm/s. Only the data within the range shown as (a) and (b) are used, to eliminate the influence of the acceleration and the deceleration.

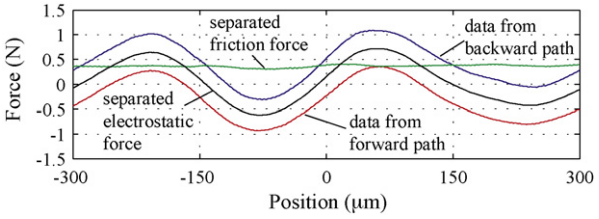


Fig. 17. An example of the separation of forces into two components: the electrostatic force and the friction force.

force. The electrostatic force can be obtained as the average of the forces from the forward path and the backward path, whereas the friction force can be calculated as half of the difference of the forces measured in the two paths. The separation was possible, because the friction force is direction dependent, whereas the electrostatic thrust force is independent. An example of the data before and after the separation is shown in Fig. 17.

The forces were measured under different frequencies, and the results are shown in Fig. 18. The electrostatic thrust forces are represented by their maximum values, and the friction forces are represented by their averaged values. The measured forces were larger than the theoretical values, for all frequencies. The results indicate that the forces become the largest at 5.6 kHz. However, the master–slave mechanism could not operate at this frequency, and the same applied to the frequencies lower than the resonance frequency. The reason for this unexpected behavior is still under investigation.

Figure 19 shows the measured forces at 5.95 kHz, with and without the inductors, plotted against the position difference of the two motors. The results show that the forces are successfully increased by the inductors. The force curves agree with the theory, and indi-

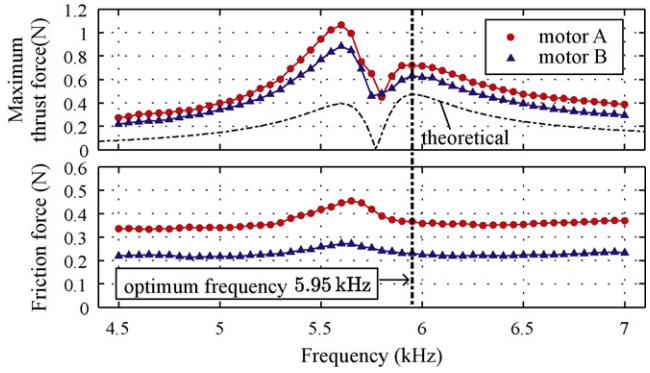


Fig. 18. Frequency dependence of the forces of the two motors. The top graph shows the maximum electrostatic thrust forces, and the bottom graph shows the averaged friction forces. The amplitude of the input voltage was 660 V.

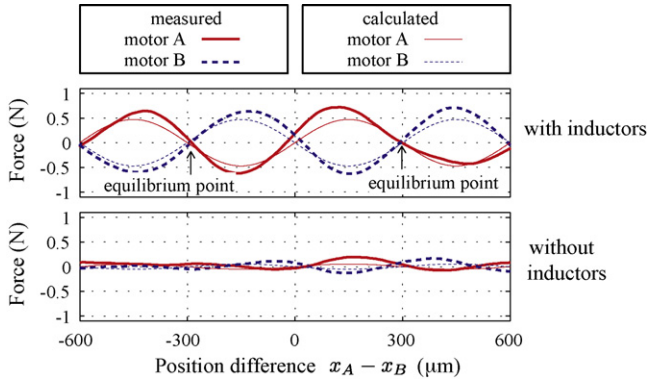


Fig. 19. Forces of the two motors plotted against the position difference. The input voltage was 660 V/5.95 kHz. The friction forces are eliminated.

cate that maximum force of about 0.6 N can be transmitted from one motor to the other.

4.4. Synchronization of two motors

The synchronization of the two motors was demonstrated. In the experiment, we chose the motor A as the master motor and the motor B as the slave motor. The motor A was forced to follow certain trajectories by the positioning stage attached to it. The motor B was set free: the positioning stage for the motor was removed from the setup. At this condition, the positions of the two sliders were measured by the laser displacement sensors.

Figure 20 shows the fastest back-and-forth pattern in which the synchronization succeeded under the optimum input voltage 660 V/5.95 kHz. The reaction force on the master motor is also shown in the figure. The maximum velocity was 20 mm/s and the maximum acceleration was 0.16 m/s^2. When the acceleration was set higher than this value, the synchronization failed: the slave motor stepped out of phase and could no longer follow the motion of the master motor. A large velocity ripple of almost 100% was

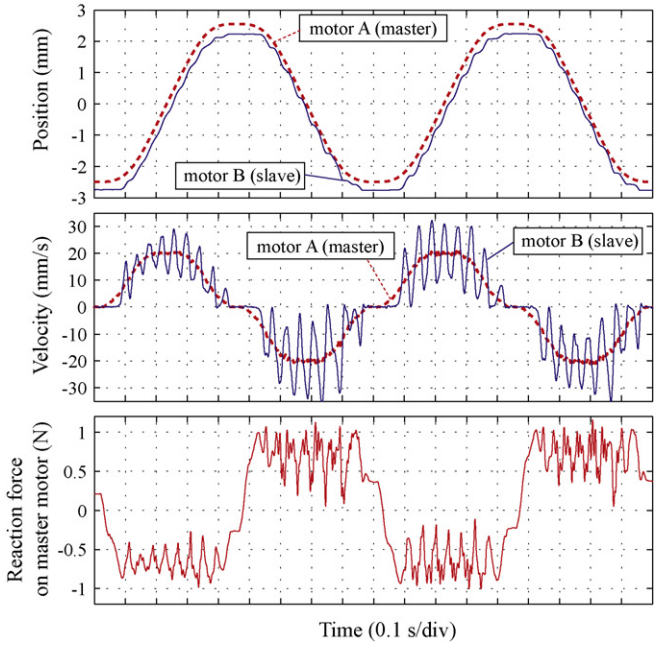


Fig. 20. Synchronization of two motors demonstrated at maximum velocity of 20 mm/s and maximum acceleration of 0.16 m/s^2. The amplitude of the input voltage was 660 V and the frequency was 5.95 kHz. The reaction force on the master motor (motor A) was also measured during the experiment.

observed during the synchronization. The ripple was transmitted to the master motor, and appeared in the reaction force. The ripple can also be observed in the position data, as a fluctuation with a period of about 600 μm, which is equivalent to three electrode pitches.

5. Discussion

The model of the electrostatic master–slave mechanism provided in this paper focuses on the simple explanation of the principles. Even though the model was idealized, it agreed well with the results of the voltage measurements, and explained the two basic characteristics: the synchronization of positions and the transmission of forces.

However, two characteristics remain unexplained due to the idealization. First is the force ripple characteristics. The ripple must be reduced in order for this mechanism to be utilized as a force transmitter. Ripple reduction methods such as the use of skewed electrodes [23] must be investigated. Second is the dynamic behavior of the mechanism. In deriving the expression of the forces, velocity dependence of the voltages were neglected, assuming mechanically quasi-static conditions. The scope of the present model in terms of velocity needs to be clarified.

In terms of fabrication, reduction of friction forces is necessary. For the prototyped master–slave mechanism, more than 30% of the maximum thrust forces were used to overcome the friction forces, which means that the dead-band of the force transmission is also ±30%. Guides capable of maintaining small gap with less friction need to be considered.

6. Conclusion

A simple master–slave mechanism comprised of two electrostatic motors is reported in this paper. The mechanism enables the synchronization of the two motor positions and the transmission of the forces from one motor to the other. The forces of the two motors are enhanced by three external inductors. The inductors cancel the self- and the parasitic capacitances of the slider electrodes and increase the slider voltages. At optimum input voltage of 660 V/5.95 kHz, the synchronization of two motors with a velocity of 20 mm/s and an acceleration of 0.16 m/s^2 was demonstrated. The maximum force that could be transmitted at this condition was 0.6 N.

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